

FOREARM MUSCLES

PLAN

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I. INTRODUCTION:

The forearm consists of a group of muscles divided into three compartments: anterior, posterior, and lateral. These muscles act by moving the hand and fingers.
The compartments are separated by the anterior and lateral intermuscular septa.

II. DESCRIPTIVE ANATOMY:

A. Anterior compartment:

Contains 8 muscles arranged in three layers.

a) Superficial layer: contains the medial epicondylar muscles.

1. Pronator teres muscle:

- Origin: two heads:
 - Humeral head: medial epicondyle.
 - Ulnar head: coronoid process
- Belly: the heads merge into a flattened belly.
- Insertion: lateral surface of the radius.
- Action: pronation and flexion of the forearm.
- Innervation: median nerve.

2. Flexor carpi radialis muscle:

- Origin: medial epicondyle.
- Belly: fusiform muscle.
- Insertion: base of the 2nd metacarpal.

- Action: wrist flexion, hand abduction.
 - Innervation: median nerve.
- 3. Palmaris longus muscle:**
- Origin: medial epicondyle.
 - Belly: slender and elongated.
 - Insertion: palmar aponeurosis.
 - Action: wrist flexion.
 - Innervation: median nerve.
- 4. Flexor carpi ulnaris muscle:**
- Origin: medial epicondyle.
 - Belly: elongated muscle stretching from the medial epicondyle of the humerus to the pisiform.
 - Insertion: pisiform.
 - Action: wrist flexion and adduction.
 - Innervation: ulnar nerve.

b) Intermediate layer:

- 5. Flexor digitorum superficialis muscle:**
- Origin:
 - Humero-ulnar head: coronoid process of the ulna and medial epicondyle of the humerus.
 - Radial head: anterior border of the radius.
 - Belly: long and thick muscle, located anterior to the deep flexors.
 - Insertion: sides of the middle phalanges of the last four fingers.
 - Action: flexion of the finger phalanges and the hand.
 - Innervation: median nerve.

c) Deep layer:

- 6. Flexor digitorum profundus muscle:**
- Origin: anterior and medial surfaces of the ulna and the interosseous membrane.
 - Belly: broad and thick, extending to the last four fingers.
 - Insertion: bases of the distal phalanges of the last four fingers.
 - Action: flexion of the finger phalanges and the hand.
 - Innervation: lateral half by the median nerve, medial half by the ulnar nerve.
- 7. Flexor pollicis longus muscle:**
- Origin: anterior surface of the radius.
 - Belly: elongated, located lateral to the deep finger flexor.
 - Insertion: base of the distal phalanx of the thumb.
 - Action: flexion of the thumb phalanges.
 - Innervation: median nerve.
- 8. Pronator quadratus muscle:**
- Origin: lower quarter of the anterior surface of the ulna.
 - Belly: flat muscle stretched between radius and ulna.
 - Insertion: lower quarter of the anterior surface of the radius.

- Action: pronation of the forearm and hand.
- Innervation: median nerve.

B. Posterior compartment:

Contains the extensor muscles, 8 in total, arranged in two layers.

a) Superficial layer: (lateral epicondylar muscles)

1. Anconeus muscle:

- Origin: lateral epicondyle of the humerus.
- Belly: triangular muscle extending to the olecranon.
- Insertion: posterior and lateral surface of the olecranon.
- Action: forearm extension.
- Innervation: radial nerve.

2. Extensor carpi ulnaris muscle:

- Origin:
 - Humeral head: lateral epicondyle.
 - Ulnar head: posterior border of the ulna.
- Belly: elongated muscle extending from the lateral epicondyle to the 5th metacarpal.
- Insertion: base of the 5th metacarpal.
- Action: wrist extension and adduction.
- Innervation: radial nerve.

3. Extensor digiti minimi muscle:

- Origin: lateral epicondyle.
- Belly: long, slender muscle.
- Insertion: bases of the 2nd and 3rd phalanges of the little finger.
- Action: extension of the little finger.
- Innervation: radial nerve.

4. Extensor digitorum muscle:

- Origin: lateral epicondyle.
- Belly: elongated muscle stretched from the lateral epicondyle to the last 4 fingers.
- Insertion: bases of the 2nd and 3rd phalanges of the last four fingers.
- Action: extension of the last four fingers.
- Innervation: radial nerve.

b) Deep layer :

5. Abductor pollicis longus muscle:

- Origin: posterior surfaces of the radius, ulna, and interosseous membrane.
- Belly: fusiform muscle extending from the posterior antebrachial region to the 1st metacarpal.
- Insertion: base of the 1st metacarpal.
- Action: abduction and extension of the thumb.
- Innervation: radial nerve.

6. Extensor pollicis brevis muscle:

- Origin: posterior surface of the radius and interosseous membrane.
- Belly: fusiform and slender muscle.
- Insertion: base of the proximal phalanx of the thumb.

- Action: extension and abduction of the thumb.
 - Innervation: radial nerve.
- 7. Extensor pollicis longus muscle:**
- Origin: posterior surface of the ulna and interosseous membrane.
 - Belly: fusiform muscle extending from the forearm posterior to the thumb.
 - Insertion: base of the distal phalanx of the thumb.
 - Action: thumb extension.
 - Innervation: radial nerve.
- 8. Extensor indicis muscle:**
- Origin: posterior surface of the ulna and interosseous membrane.
 - Belly: fusiform and slender muscle.
 - Insertion: bases of the middle and distal phalanges of the index finger.
 - Action: index finger extension.
 - Innervation: radial nerve.

Note: The two thumb extensors diverge at the lateral border of the wrist to form the anatomical snuffbox.

C. Lateral compartment:

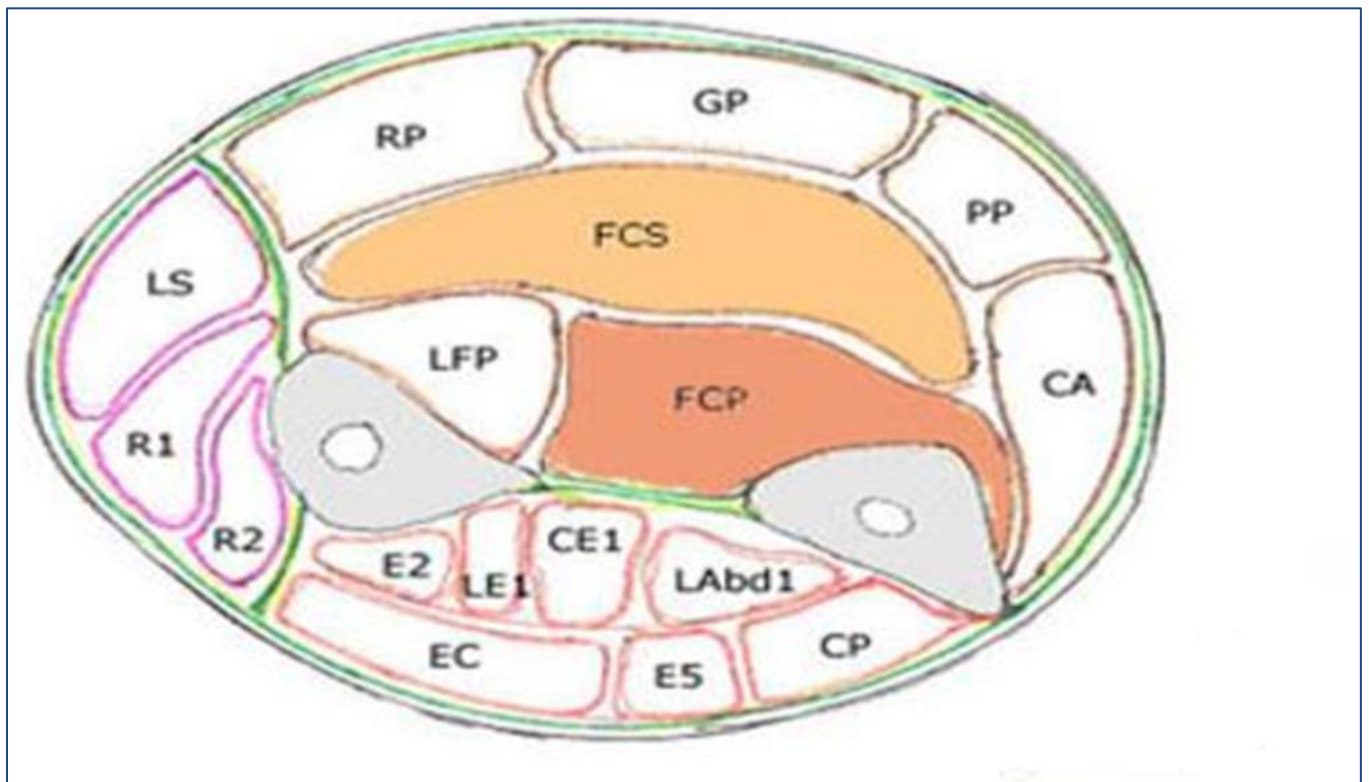
Contains the supinators and radial wrist extensors.

- 1. Brachioradialis muscle:**
 - Origin: lateral border of the humerus and lateral epicondyle.
 - Belly: thick and elongated muscle.
 - Insertion: lateral side of the styloid process of the radius.
 - Action: forearm flexion, supination after prior pronation.
 - Innervation: radial nerve.
- 2. Extensor carpi radialis longus muscle:**
 - Origin: lateral epicondyle.
 - Belly: elongated muscle extending from the humerus to the 2nd metacarpal.
 - Insertion: base of the 2nd metacarpal.
 - Action: wrist extension and abduction.
 - Innervation: radial nerve.
- 3. Extensor carpi radialis brevis muscle:**
 - Origin: lateral epicondyle.
 - Belly: elongated muscle extending from the humerus to the 3rd metacarpal.
 - Insertion: base of the 3rd metacarpal.
 - Action: wrist extension and abduction.
 - Innervation: radial nerve.
- 4. Supinator muscle:**
 - Origin:
 - Humeral (superficial) head: lateral epicondyle.
 - Ulnar (deep) head: posterolateral surface of the ulna.
 - Belly: large muscle located in the upper third of two bones of the forearm.

- Insertion: lateral surface of the radius.
- Action: forearm supination.
- Innervation: radial nerve.

References:

- Hammoudi SS. Le cours d'anatomie (descriptive topographique et fonctionnelle).Appareil locomoteur I. Membre supérieur.(traduit en anglais).
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Horizontal cut at the level of the middle 1/3 of the forearm

